

JEE Main - 2 | JEE - 2023 (Gen 1 & 2)

Date: 06/07/2021

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 9.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

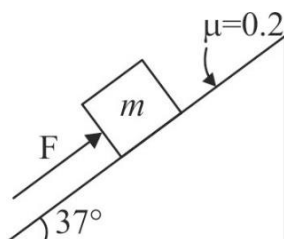
PART - I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The velocity of the particle moving along the X-axis is given as a function of its position by: $v(x) = 4 - 3x$. The acceleration of the particle as a function of its position is given by:

- (A) $a(x) = 3(4 - 3x)$ (B) $a(x) = 3(3x - 4)$
(C) $a(x) = 6(4 - 3x)$ (D) $a(x) = 6(3x - 4)$

2. A block of mass 5 kg is being held on a fixed, rough inclined plane ($\mu = 0.2$) of angle of inclination 37° by applying a force F parallel to the plane as shown in the figure. What should be the minimum value of F to hold the block stationary? ($\cos 37^\circ = 0.8$, $\sin 37^\circ = 0.6$, $g = 10 \text{ m/s}^2$)



- (A) 30 N (B) 22 N (C) 8 N (D) 5 N

3. A particle is projected from the ground at $t = 0$ with velocity 50 m/s at an angle 30° with the horizontal. Then, at $t = 2 \text{ s}$: ($g = 10 \text{ m/s}^2$)

- (I) the velocity of the particle is directed at an angle $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ with the horizontal

- (II) the speed of the particle is $10\sqrt{19} \text{ m/s}$

Which of these options is correct?

- (A) Both (I) and (II) are correct (B) (I) is correct and (II) is incorrect
(C) (II) is correct and (I) is incorrect (D) Both (I) and (II) are incorrect

4. Two particles A and B initially at rest a distance 200 m apart simultaneously start moving directly towards each other at constant acceleration 3 m/s^2 and 1 m/s^2 respectively. At the instant they meet, their velocities (in m/s) are respectively:

- (A) 15, 5 (B) 30, 10 (C) $15\sqrt{2}$, $5\sqrt{2}$ (D) $30\sqrt{2}$, $10\sqrt{2}$

5. The position of a particle moving on the X-axis is given as a function of time by: $x(t) = t^3 - 2t^2 + 4t - 5$. Here, x is in metres, and t is in seconds. At $t = 3 \text{ s}$, the velocity and acceleration of the particle are respectively:

- (A) 19 m/s, 14 m/s^2 (B) 1 m/s, 2 m/s^2
(C) 19 m/s, 5 m/s^2 (D) 14 m/s, 18 m/s^2

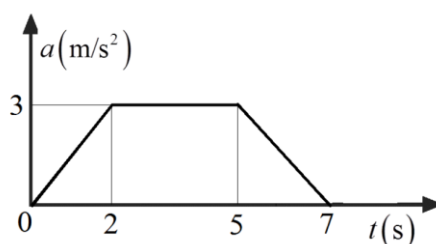
6. A person running along a straight track with a uniform velocity $\vec{v} = v\hat{i}$ feels that the rain is falling vertically downwards along $-\hat{j}$. If he doubles his speed, he sees the rain coming at an angle α with the horizontal. The velocity of the rain with respect to the ground is:

(A) $-v\hat{i} - \frac{v}{\tan \alpha} \hat{j}$ (B) $-v\hat{i} + v \tan \alpha \hat{j}$ (C) $v\hat{i} - \frac{v}{\tan \alpha} \hat{j}$ (D) $v\hat{i} - v \tan \alpha \hat{j}$

7. A car moving in a straight line travels the first one-third distance at a constant velocity 8 m/s, the next one-third distance at constant velocity 12 m/s, and the last one-third distance at constant velocity 20 m/s. The average velocity of the car for the entire journey is:

(A) 11.19 m/s (B) 11.61 m/s (C) 12.28 m/s (D) 12.94 m/s

8. A car starts from rest at $t=0$ and moves in a straight line with a time-varying acceleration depicted in the graph. In the interval $2 \text{ s} \leq t \leq 5 \text{ s}$, the velocity of the car follows the relation:



(A) $v(t) = 3t \text{ m/s}$ (B) $v(t) = (3t - 3) \text{ m/s}$
(C) $v(t) = (3t - 6) \text{ m/s}$ (D) $v(t) = 3 \text{ m/s}$

9. The velocity of a particle moving along the X-axis is given as a function of time by: $v = 8\cos(2t)$. Here, v is in m/s and t is in seconds. If the position of the particle at $t=0$ is $x=10 \text{ m}$, then its position (in metres) at $t = \frac{\pi}{6} \text{ s}$ is:

(A) $x = 4(\sqrt{2} + 1)$ (B) $x = 4(\sqrt{2} - 1)$ (C) $x = 2(5 + \sqrt{3})$ (D) $x = 2(5 - \sqrt{3})$

10. If $\vec{P} = 2\hat{i} + \hat{j} + 2\hat{k}$, $\vec{Q} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{P} \times \vec{Q}$ is parallel to the vector $\alpha\hat{i} + \beta\hat{j} + 6\hat{k}$, then values of α and β are respectively:

(A) $-2, -8$ (B) $3, 4$ (C) $-3, -4$ (D) $2, 8$

11. Starting from rest at $t=0$, a particle moves in a straight line with a constant acceleration. If for an integer n , S_n denotes the distance travelled by the particle between $t = (n-1) \text{ sec}$ and $t = n \text{ sec}$, the ratio $\frac{S_2}{S_4}$ is:

(A) $\frac{1}{3}$ (B) $\frac{2}{5}$ (C) $\frac{3}{7}$ (D) $\frac{4}{9}$

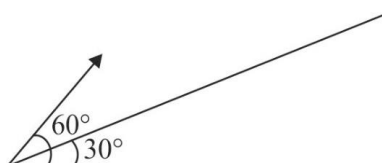
12. Starting from rest, a particle moves along a circle of radius 9 m with a constant tangential acceleration 3 m/s^2 . At $t = 2 \text{ s}$, the total acceleration of the particle makes an angle θ with its tangential acceleration. Then, θ is equal to:

(A) $\tan^{-1}(2)$ (B) $\tan^{-1}\left(\frac{5}{7}\right)$ (C) $\tan^{-1}\left(\frac{1}{3}\right)$ (D) $\tan^{-1}\left(\frac{4}{3}\right)$

13. The X and Y coordinates of a particle moving in the X - Y plane are given as functions of time by: $x(t) = 4 - 2t$ and $y(t) = 3t^2$. The equation of trajectory of the particle is:

(A) $y = 3(2 - x)^2$ (B) $y = 3(4 - x)^2$ (C) $y = \frac{3}{4}(2 - x)^2$ (D) $y = \frac{3}{4}(4 - x)^2$

14. A particle is projected at time $t = 0$ with velocity 20 m/s at an angle of 60° with the horizontal, from the foot of an inclined plane. The inclined plane makes an angle 30° with the horizontal. The time instant at which the velocity of the projectile becomes parallel to the incline is: ($g = 10 \text{ m/s}^2$)

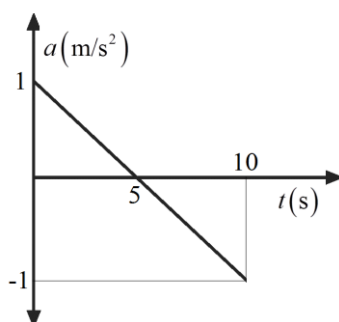


(A) $t = \frac{1}{\sqrt{3}} \text{ s}$ (B) $t = \frac{2}{\sqrt{3}} \text{ s}$ (C) $t = 1 \text{ s}$ (D) $t = \frac{3}{2} \text{ s}$

15. A particle moves along a circle with speed $v = at$, where $a = 0.2 \text{ m/s}^2$. At the moment the particle has covered $\left(\frac{1}{8}\right)$ th of the circle after the beginning of motion, the magnitude of its radial acceleration (in m/s^2) is:

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{5}$ (C) $\frac{\pi}{10}$ (D) $\frac{\pi}{16}$

16. A particle initially at rest moves in a straight line with a time-varying acceleration depicted in the graph. The velocity of the particle at $t = 8 \text{ s}$ is:



(A) 2.0 m/s (B) 1.6 m/s (C) 1.2 m/s (D) 0.8 m/s

17. The velocity of a particle moving along a straight line is given as a function of time as: $v(t) = 6t^2 + 5$. Here, v is in m/s and t is in seconds. The average velocity (in m/s) of the particle between the instants $t = 1 \text{ s}$ and $t = 3 \text{ s}$ is:

(A) 31 (B) 33.5 (C) 34.5 (D) 37

18. A particle projected from the ground at an angle 30° with the horizontal has maximum height H_0 and range R_0 . If the particle was projected with half the initial velocity at an angle 45° with the horizontal, its maximum height and range respectively would be:

(A) $\frac{H_0}{4}, \frac{R_0}{2}$ (B) $\frac{H_0}{2}, \frac{R_0}{2\sqrt{3}}$ (C) $\frac{H_0}{2}, \frac{R_0}{\sqrt{3}}$ (D) $\frac{H_0}{4}, \frac{2R_0}{\sqrt{3}}$

19. A particle, initially at rest at the origin, starts moving in the X-Y plane at $t=0$ with a constant acceleration. If the position of the particle at $t=2$ s is $(-2, 4)$, its velocity at this instant is:
- (A) $-\hat{i} + 2\hat{j}$ (B) $-2\hat{i} + 4\hat{j}$ (C) $-4\hat{i} + 8\hat{j}$ (D) $\hat{i} - 2\hat{j}$
20. A particle is projected from the ground at an angle 45° with the horizontal. If it lands on the ground after time 4 s, the horizontal range (in metres) of the particle is: ($g = 10 \text{ m/s}^2$)
- (A) 20 (B) 40 (C) 80 (D) 120
-

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. A ball is released from rest from a height h above the ground at $t = 0$. Every time the ball collides with the ground, its velocity becomes half of its value before the collision. If the ball has its second collision with the ground after being released at $t = \left(N \left(\frac{h}{g} \right) \right)^{1/2}$, then N is _____.
22. The position of a particle moving on the X -axis is given as a function of time by: $x(t) = 5 - 6t + t^2$. Here, x is in metres, and t is in seconds. The distance travelled (in metres) by the particle between $t = 2$ s and $t = 4$ s is _____.
23. A swimmer crosses a river of width d flowing at velocity v_0 . The swimmer's velocity in still water is $\frac{2v_0}{3}$. While swimming, he orients his effort in a direction at an angle 127° with the river flow. Upon reaching the other bank, he finds he has drifted a distance L in the direction of flow of the river. If $L = \left(1 + \frac{1}{N} \right) d$, then N is _____. ($\cos 37^\circ = 0.8$, $\sin 37^\circ = 0.6$)
24. A particle moves in a straight line with deceleration $\frac{3}{4}\sqrt{v}$, where v is its instantaneous velocity. If at a particular instant, the particle has velocity 2.25 m/s, the time taken (in seconds) after this instant for the particle to come to rest is _____.
25. An airplane is flying horizontally at constant velocity 60 m/s at a height 540 m. A box is released from the airplane. If the velocity (in m/s) with which the box lands on the ground is $20X$, then X is _____. ($g = 10 \text{ m/s}^2$)

SPACE FOR ROUGH WORK

PART - II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- The third emission line in the atomic spectrum of hydrogen in the Balmer series appears at (R=Rydberg constant) :
 (A) $\frac{9R}{400} \text{ cm}^{-1}$ (B) $\frac{7R}{144} \text{ cm}^{-1}$ (C) $\frac{5R}{36} \text{ cm}^{-1}$ (D) $\frac{21R}{100} \text{ cm}^{-1}$
- If λ_0 is the threshold wavelength for photoelectric emission, λ wavelength of light falling on the surface of metal, and m, mass of electron, then de Broglie wavelength of emitted electron is:
 (A) $\left[\frac{h(\lambda\lambda_0)}{2mc(\lambda_0 - \lambda)} \right]^{\frac{1}{2}}$ (B) $\left[\frac{h(\lambda_0 - \lambda)}{2mc\lambda\lambda_0} \right]^{\frac{1}{2}}$
 (C) $\left[\frac{h(\lambda - \lambda_0)}{2mc\lambda\lambda_0} \right]^{\frac{1}{2}}$ (D) $\left[\frac{h\lambda\lambda_0}{2mc} \right]^{\frac{1}{2}}$
- The maximum number of electrons in a subshell having the same value of spin quantum number is given by:
 (A) $\ell + 2$ (B) $2\ell + 1$ (C) $2(2\ell + 1)$ (D) None
- Give the correct order of initials T (true) of F (false) for following statements.
 (i) The ratio of the radii of the first three Bohr orbit of hydrogen atom is 1 : 8 : 27
 (ii) The ratio of velocity of an electron in 2nd and 3rd orbit of hydrogen atom is 3 : 2
 (iii) The frequency of a green light is 6×10^{14} Hz, then its wavelength is 500 nm
 (iv) The equivalent mass of H_3PO_4 when it combines with NaOH using the equation

$$\text{H}_3\text{PO}_4 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{HPO}_4 + 2\text{H}_2\text{O}$$
 is 49 g. (Given: Molar mass of H_3PO_4 is 98 g/mole)
 (A) FFTT (B) FTFT (C) TFTF (D) FTTT
- The ratio of time periods in first and third orbits of hydrogen atom is:
 (A) 1 : 3 (B) 1 : 4 (C) 1 : 27 (D) 1 : 9
- Which of the following orbitals has two spherical nodes?
 (A) 2s (B) 4p (C) 3d (D) 5f
- A particle initially at rest having charge q coulomb. & mass m kg is accelerated by potential difference of V volts. What would be its K.E & de Broglie wavelength respectively after acceleration.
 (A) $qV, \frac{h}{\sqrt{2qVm}}$ (B) $\frac{h}{\sqrt{2qVm}}, qV$ (C) $qV, \frac{h}{mV}$ (D) $\frac{h}{mV}, qV$
- If radiation corresponding to first line of "Balmer series" of He^+ ion knocked out electron from 1st excited state of H atom, the kinetic energy of ejected electron from H atom would be (eV) :
 (A) 4.155 eV (B) 8.310 eV (C) 2.515 eV (D) 5.550 eV

9. If the ionization energy of He^+ is 19.6×10^{-18} J per atom then the energy of Li^{2+} ion in the second stationary state is:
 (A) -4.9×10^{-18} J (B) -44.1×10^{-18} J
 (C) -11.025×10^{-18} J (D) None of these
10. If the radius of 3rd Bohr's orbit in H is x, then radius of 4th orbit of Li^{2+} ion would be:
 (A) $\frac{27}{16}x$ (B) $\frac{16}{27}x$ (C) $\frac{9}{16}x$ (D) None of these
11. Light of wavelength λ strike a metal surface with intensity 'x' and the metal emits 'y' electrons per second of maximum energy 'z'. What will happen to 'y' and 'z' if 'x' is halved.
 (A) y will be halved (B) y will be doubled
 (C) z will be doubled (D) z will halved
12. Which electronic transition in Li^{+2} will emit photon of same energy as that emitted due to transition from 5th excited state to 1st excited state in He^+ .
 (A) $6 \rightarrow 2$ (B) $3 \rightarrow 1$ (C) $9 \rightarrow 3$ (D) $2 \rightarrow 6$
13. Three photons originating from excited atomic hydrogen atoms are found to have energies of 0.66eV, 1.89eV, 2.55eV respectively. The minimum number of atoms that must be present are :
 (A) One atom (B) Two atoms (C) Three atoms (D) Can't say
14. Calculate the mass of sucrose $\text{C}_{12}\text{H}_{22}\text{O}_{11}(\text{s})$ produced by mixing 78g of C(s), 11g of $\text{H}_2(\text{g})$ & 67.2 litre of $\text{O}_2(\text{g})$ at STP according to given reaction (unbalanced) ? (Given: Molar mass of sucrose = 342 g/mole and atomic mass of H = 1 and C = 12)

$$\text{C}(\text{s}) + \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow \text{C}_{12}\text{H}_{22}\text{O}_{11}(\text{s})$$

 (A) 171g (B) 155.4g (C) 185.25g (D) None of these
15. Aspartame, an artificial sweetener contains 9.52 weight % of nitrogen. There are two nitrogen atoms per molecule. What is molecular weight of aspartame ? (Given: Atomic mass of N = 14)
 (A) 147 (B) 294 (C) 588 (D) 266
16. The total number of neutrons present in 10g of D_2O (D is ${}^2_1\text{H}$) are : (Given: Atomic mass of O = 16)
 (A) $2.5 N_A$ (B) $5.0 N_A$ (C) $10.0 N_A$ (D) None of these
17. Chlorine gas can be produced by the reaction of $\text{HCl}(\text{aq})$ with $\text{MnO}_2(\text{s})$. Only $\text{MnCl}_2(\text{s})$ and $\text{H}_2\text{O}(\ell)$ are the by products. What volume of $\text{Cl}_2(\text{g})$ (in litre) of density 2.84 g/L will be produced from the reaction of 400mL of 0.1 M $\text{HCl}(\text{aq})$ with an excess of MnO_2 ? (Atomic weight of Cl = 35.5)
 (A) 1 (B) 0.5 (C) 0.25 (D) 2
18. A mixture of hydrocarbon C_2H_2 , C_2H_4 & CH_4 in mole ratio of 2 : 1 : 2 is burnt completely in the presence of air containing 80% N_2 & 20% O_2 by mole. The mass of air required for the complete combustion of the one gm of mixture is : (Given: Atomic mass of H = 1, C = 12, N = 14, O = 16)
 (A) $\frac{1728}{112}$ (B) $\frac{1528}{73}$ (C) $\frac{1920}{120}$ (D) $\frac{112}{1728}$

19. 100 ml of 2.45% (w/v) H_2SO_4 solution is mixed with 200 ml of 7% (w/w) H_2SO_4 solution (density = 1.4 gm / ml) and the mixture is diluted to 500 ml. What is the molarity of the diluted solution ? (Atomic weight of S = 32 g)
- (A) 0.25 M (B) 0.1 M (C) 0.75 M (D) 0.45 M
20. Which of the options is/are incorrect regarding the order of frequency of Electromagnetic Radiation ?
- (A) Radiowaves < Blue light < Ultraviolet (B) Green light < Yellow light < Infra red
- (C) Indigo light < X-rays < Gamma rays (D) Microwaves < Orange light < Ultraviolet
-

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. Radiation corresponding to the transition $n=4$ to $n=2$ in hydrogen atoms falls on a certain metal (work function = 1.55 eV). The maximum kinetic energy (in eV) of the photoelectrons will be:
22. 14 g of a sample containing $(\text{NH}_4)_2\text{SO}_4$ & inert impurity (X) boiled with excess of NaOH and the $\text{NH}_3(\text{g})$ so produced was absorbed in 200 ml of 0.5 M H_2SO_4 solution in another vessel. The unreacted H_2SO_4 required 400 ml of 0.4 M solution of NaOH for exact neutralization. Calculate weight % of nitrogen in given sample. (Atomic weight of N = 14 g)
23. 15gm mixture of SiO_2 and Fe_2O_3 on very strong heating leave a residue weighing 14.96gm. The reaction responsible for loss of weight is :
- $$\text{Fe}_2\text{O}_3(\text{s}) \longrightarrow \text{Fe}_3\text{O}_4(\text{s}) + \text{O}_2(\text{g})$$
- What is percentage by mass of Fe_2O_3 in original sample ? (Atomic weight of Fe = 56 g)
24. H_2SO_4 solution (80% by weight and density 1.75 g/ml is used to prepare 1 litre of $\frac{1}{14}$ M $\text{H}_2\text{SO}_4(\text{aq})$. The volume of H_2SO_4 solution (original in ml) which must be used is :
25. An unknown metal reacts with excess chlorine to give the metal chloride MCl_2 . When 2.375gm of metal chloride is dissolved in water and passed through an anion-exchange column packed with hydroxide ion, the solution requires 100ml of 0.5 M HCl for neutralization. Calculation atomic mass (in gm/mole) of unknown metal. Fill your answer by dividing it by 4 :

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

- The number of four digit numbers greater than 4000 that can be formed using the digits 1, 2, 3, 4, 5, 6 (repetition of digits is allowed) is equal to:
(A) 1296 (B) 648 (C) 180 (D) 360
- The sum of the series $1 + 4 \times 3 + 9 \times 5 + 16 \times 7 + 25 \times 9 + \dots$ upto 15 terms is:
(A) 30040 (B) 28800 (C) 25660 (D) 27560
- A team of 11 members has to be formed from 7 males and 8 females. If 'm' is the number of ways team is formed with atleast 6 males and 'n' is the number of ways team is formed with atleast 6 females, then:
(A) $m + n = 1362$ (B) $m + n = 1265$ (C) $n - m = 441$ (D) $n = 2m$
- Let a, b, c be in G.P. with common ratio 'r' where $a \neq 0$ and $0 < r < 1$. If $8a, 7b, 3c$ are the first three terms of an A.P., then the fourth term of this A.P. is:
(A) $\frac{10}{3}a$ (B) $\frac{14}{3}a$ (C) $-2a$ (D) $-3a$
- The number of 4 letter words (with or without meaning) that can be formed from the 9 letters of the word 'PRACTICAL' is:
(A) 1206 (B) 1566 (C) 66 (D) 1584
- $\sum_{k=1}^5 (1 + 3 + 5 + 7 + \dots (2k - 1))$ is equal to:
(A) 25 (B) 15 (C) 45 (D) 55
- The product $\left(\frac{1}{3^4}\right)\left(\frac{1}{9^{16}}\right)\left(\frac{1}{27^{48}}\right) \dots$ infinite terms is equal to:
(A) 3 (B) $\frac{1}{3^2}$ (C) $\frac{1}{3^4}$ (D) None of these
- The number of diagonals of n sided polygon is four times the number of sides of the same polygon. The total number of triangles that can be formed using the vertices of given polygon is equal to:
(A) 990 (B) 165 (C) 330 (D) 660
- The total number of positive integral solutions (x, y, z) such that $xyz = 378$ is:
(A) 90 (B) 60 (C) 45 (D) 30
- The sum of the series: $(1^2 + 2^2 + 1 \times 2) + (2^2 + 3^2 + 2 \times 3) + (3^2 + 4^2 + 3 \times 4) + \dots (19^2 + 20^2 + 19 \times 20)$ is equal to:
(A) 7999 (B) 8000 (C) 8001 (D) 4000
- There are 10 chairs in a row. If 3 persons wish to sit on these chairs such that there is no empty chair between any of the three persons then total number of ways of doing this is equal to:
(A) $({}^{10}C_3)(3!)$ (B) $({}^8C_3) \times 3!$ (C) 48 (D) 60

12. Letters of the word 'HISTORY' are arranged in all possible ways to form 7 letters words without repetition. These words are now arranged in a format as of dictionary; rank of 'HISTORY' in this dictionary is:
(A) 60 **(B)** 61 **(C)** 62 **(D)** 59
13. The number of ways in which the word 'COPYRIGHT' can be written so that vowels occupy same position as before and no consonant occupies its original position is equal to:
(A) 2580 **(B)** 1850 **(C)** 1784 **(D)** 1854
14. If a, b, c are distinct positive numbers in Harmonic progression, then:
(A) $\frac{1}{a^2}, \frac{1}{b^2}, \frac{1}{c^2}$ are in A.P. **(B)** $\frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ are in A.P.
(C) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in A.P. **(D)** $\frac{a-b}{b-c} = \frac{c}{a}$
15. The number of positive divisors of N where $N = (2^a)(3^b)(4^c)(5^d)(6^e)$ is equal to: (a, b, c, d and e are all positive integers)
(A) $(a+1)(b+1)(c+1)(d+1)(e+1) - 2$ **(B)** $(a+2c+1)(b+e+1)(d+1) - 2$
(C) $(a+2c+e+1)(b+e+1)(d+1) - 2$ **(D)** $(a+2c+e+1)(b+e+1)(d+1)$
16. $\sum_{r=1}^{49} \sqrt{r - \sqrt{r^2 - 1}}$:
(A) $5 + \sqrt{2}$ **(B)** $5 + 2\sqrt{2}$ **(C)** $5 - \sqrt{2}$ **(D)** $5 + 3\sqrt{2}$
17. A ring can be worn in any one of the four fingers of a hand and each finger can accommodate any number of rings. The total number of ways of putting 6 different rings on 4 fingers is equal to:
(A) 6^4 **(B)** 4^6 **(C)** $6!$ **(D)** $4!$
18. The total number of ways of selecting two different natural numbers from 1 to 30 such that their sum is divisible by three is equal to: (both 1 and 30 included)
(A) 360 **(B)** 145 **(C)** 150 **(D)** 720
19. 10 arithmetic means are inserted between 3 and 168, then the sum of first two arithmetic means is equal to:
(A) 71 **(B)** 41 **(C)** 51 **(D)** 61
20. The number of ways of arranging 'n' different objects in straight line is 'x' and in a circular loop is 'y'. If $x + y = (19)(17!)$, then n is equal to:
(A) 18 **(B)** 19 **(C)** 17 **(D)** 20

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. The number of five digit numbers with distinct digits and 3 at units place is $(448)k$, then k is equal to _____.
22. $\frac{1}{16}$, a and b are in G.P. and $\frac{1}{a}$, $\frac{1}{b}$ and $-\frac{16}{9}$ be in A.P. where $a < \frac{1}{3}$, then $4(a+b)$ is equal to _____.
23. $x = 1 + \frac{3}{5} + \frac{5}{25} + \frac{7}{125} + \dots$ infinite terms. If $x = \frac{a}{b}$ where a and b are positive co-prime integers, then $(a-b)$ is equal to _____.
24. Let p, q, r are positive real numbers such that $27pqr \geq (p+q+r)^3$ and $3p+4q+5r=12$, then $3p^3+4q^4+r^5$ is equal to _____.
25. α and β are distinct real roots of the equation $8x^2-10x+3=0$. $S_1 = \alpha + \alpha^2 + \alpha^3 + \dots$ infinite terms and $S_2 = \beta + \beta^2 + \beta^3 + \dots$ infinite terms, then $S_1 + S_2$ is equal to _____.

SPACE FOR ROUGH WORK